

Decimal to IEEE 754 32-bit, and back

Name: _____ Goes with the October 5 lecture. Work alone or in pairs. One step per box; each step uses the one before it.

The whole chain: decimal $\rightarrow$ binary $\rightarrow$ normalized $1.xxx \times 2^e \rightarrow$ sign | exponent | mantissa $\rightarrow$ 32 bits

Sign (bit 31): 0 positive, 1 negative. **Exponent** (bits 30-23): stored as $e + 127$ (the bias).

Mantissa (bits 22-0): the bits after the leading 1; the 1 is hidden, not stored.

Decode: $\text{value} = (-1)^{\text{sign}} \times 1.\text{mantissa} \times 2^{(\text{stored exponent} - 127)}$

A. Store 12.75 as a 32-bit float

1. Split it: $12.75 = \underline{\hspace{1cm}} + \underline{\hspace{1cm}}$
2. Whole part to binary. Divide by 2 until the quotient is 0; read the remainders **last to first**.

Divide	Quotient	Remainder
$12 \div 2$	6	0

$$12 = \underline{\hspace{2cm}}_2$$

Fraction part to binary. Double, copy the whole-number bit, keep the fraction; stop when Keep is 0. Read the bits **first to last**.

Double	Bit	Keep
$0.75 \times 2 =$		

$$0.75 = 0.\underline{\hspace{1.5cm}}_2$$

3. Join the two parts: $12.75 = \underline{\hspace{2cm}}_2$
4. Normalize: move the binary point until a single 1 is left of it.
Your step 3 number = $1.\underline{\hspace{2cm}}$ $\times 2\text{---}$
The point moved places to the .
5. Sign bit: because 12.75 is
6. Stored exponent: + 127 = = ₂ (8 bits)
7. Mantissa: copy the bits after the leading 1, then pad with 0s **on the right** to 23 bits.
8. Write all 32 bits in the boxes.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0					
sign	exponent (8 bits)								mantissa (23 bits)																											

B. A stored float, back to decimal

A program stored this 32-bit float, shown in hex: C1A40000. What decimal number is it?

0	1	2	3	4	5	6	7
0000	0001	0010	0011	0100	0101	0110	0111
8	9	A	B	C	D	E	F
1000	1001	1010	1011	1100	1101	1110	1111

1. Write each hex digit as its 4 bits, under the digit.

C				1				A				4				0				0				0				0			
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
sign	exponent (8 bits)							mantissa (23 bits)																							

2. Sign bit: _____ so the number is _____
3. Exponent bits: _____₂ = _____ True exponent: _____ – 127 = _____
4. Mantissa: write the bits up to the last 1 (the rest are padding zeros):

5. Put back the hidden 1: 1. _____ × 2^{_____}
6. Move the binary point by the exponent: _____₂ (which way? _____)
7. Convert to decimal with place values, then apply the sign:

place	32	16	8	4	2	1	.	1/2	1/4	1/8
your bits							.			

C1A40000 = _____

Check yourself: encode your answer back to 32 bits the way you did in Part A. You should get C1A40000.